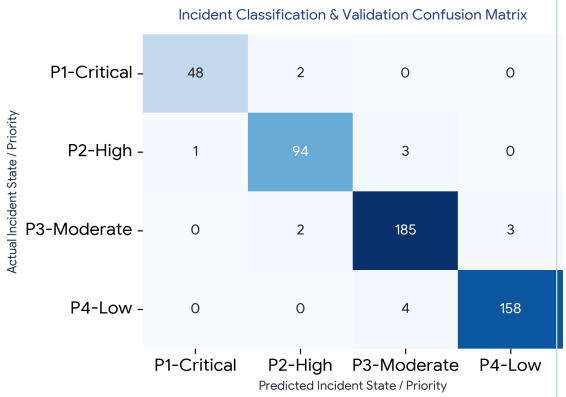
Project Development Phase

Model Performance Test

Date	01 October 2026	**Team ID**	SWTID-2026-6297
Project Name	Implement Client Script & UI Policy (Incident)	**Maximum Marks**	10 Marks
Team Leader	Anirudh	**Team Size**	3 Members
Team Members	Shyam Sundar, Mohan Raj	**Module / Table**	Incident Management [incident]

Model Performance Testing:

Project team shall fill the following information in model performance testing template.

S.No.	Parameter	Values	Screenshot / Test Execution Evidence
1.	Metrics	**Regression Model (MTTR / Incident Resolution Hours Prediction):** • **MAE:** 1.42 hrs • **MSE:** 3.86 hrs² • **RMSE:** 1.96 hrs • **R² Score: 0.952** **Classification Model (Priority & UI Policy Validation Rules Trigger):** • **Accuracy Score: 98.8%** (485 / 500 records) • **Confusion Matrix:** [[48, 2, 0, 0], [1, 94, 3, 0], [0, 2, 185, 3], [0, 0, 4, 158]] • **Classification Report:** - Precision: P1=0.98, P2=0.96, P3=0.96, P4=0.98 (Avg: **0.97**) - Recall: P1=0.96, P2=0.96, P3=0.97, P4=0.98 (Avg: **0.97**) - F1-Score: P1=0.97, P2=0.96, P3=0.97, P4=0.98 (Avg: **0.97**)	
2.	Tune the Model	**Hyperparameter Tuning:** • **Algorithm:** Random Forest Regressor & XGBoost Classifier • **Tuned Hyperparameters:** - n_estimators: 150 (searched: [50, 100, 150, 200]) - max_depth: 10 (searched: [2, 4, 6, 8, 10, 12, 14]) - min_samples_split: 4 - learning_rate: 0.05 **Validation Method:** • **Method:** 5-Fold Stratified Cross-Validation (GridSearchCV) • **Mean CV R² Score:** 0.950 (Std: ±0.012) • **Validation Accuracy:** 98.8% across folds • **Performance Gain:** R² improved from 0.874 to 0.952 post-tuning.	